

COMPUTER VISION IG.2405 – PROJECT 2026

This document describes the project to be carried out by teams of up to 4 students: specifications, instructions, challenge, deliverables, and evaluation.

1 Introduction

The company DeepForm is launching a project to commercialize a tool for automatic reading of semi-structured exam forms. “Semi-structured” means that the form contains well-defined elements but with variable layout, graphical elements such as checkboxes, as well as fields to be filled in with handwritten numbers or letters. Moreover, there is no separate answer sheet: questions and answer areas are on the same document.

2 Scénario

2.1 Exam

A paper copy of the file `EXAM_FORMXX.pdf` is distributed to each student. Each student fills in the identification page (page 1), takes a photo of it, and uploads it to the designated website. Students then read the instructions and answer the questions. They submit the paper form at the end of the exam.

2.2 Data processing

The professor receives all images of the first pages (one per student), stored in the folder: `EXAM_FORMXX_PRESENCES`.

The professor scans all paper forms and creates one PDF per form: `EXAM_FORMXX_0001.pdf`, `EXAM_FORMXX_0002.pdf`, etc.. These files are stored in the folder `EXAM_FORMXX_PDF`.

2.3 Automatic grading

The application has 3 objectives:

1. Validate attendance and authenticate signatures using first-page photos.
2. Authenticate each scanned form (exam and student) and automatically read it.
3. Assign a grade.

Thus, the professor obtains each student’s grade, validated if:

- the signature ID matches the student ID (photo and PDF),
- the correct form is used.

3 Project

To carry out this project, the company DeepForm is asking Isep students in the IG.2405 module to work on attendance validation from the first page images, and on the automated reading of scanned PDF forms.

3.1 Input data

3.1.1 General student data

- `STUDENT_CLASS_SIGNATURES`: database of student signatures.

3.1.2 Exam data

- `EXAM_FORMXX_PRESENCES`: directory of the first-page images.

- **EXAM_FORMXX_PDF**: directory of the scanned PDF forms, **EXAM_FORMXX_0001.pdf**, **EXAM_FORMXX_0002.pdf**, ..., **EXAM_FORMXX_0068.pdf**, if 68 students did the exam.

3.2 General structure

The diagram below shows the main steps to perform attendance validation and automatic form reading.

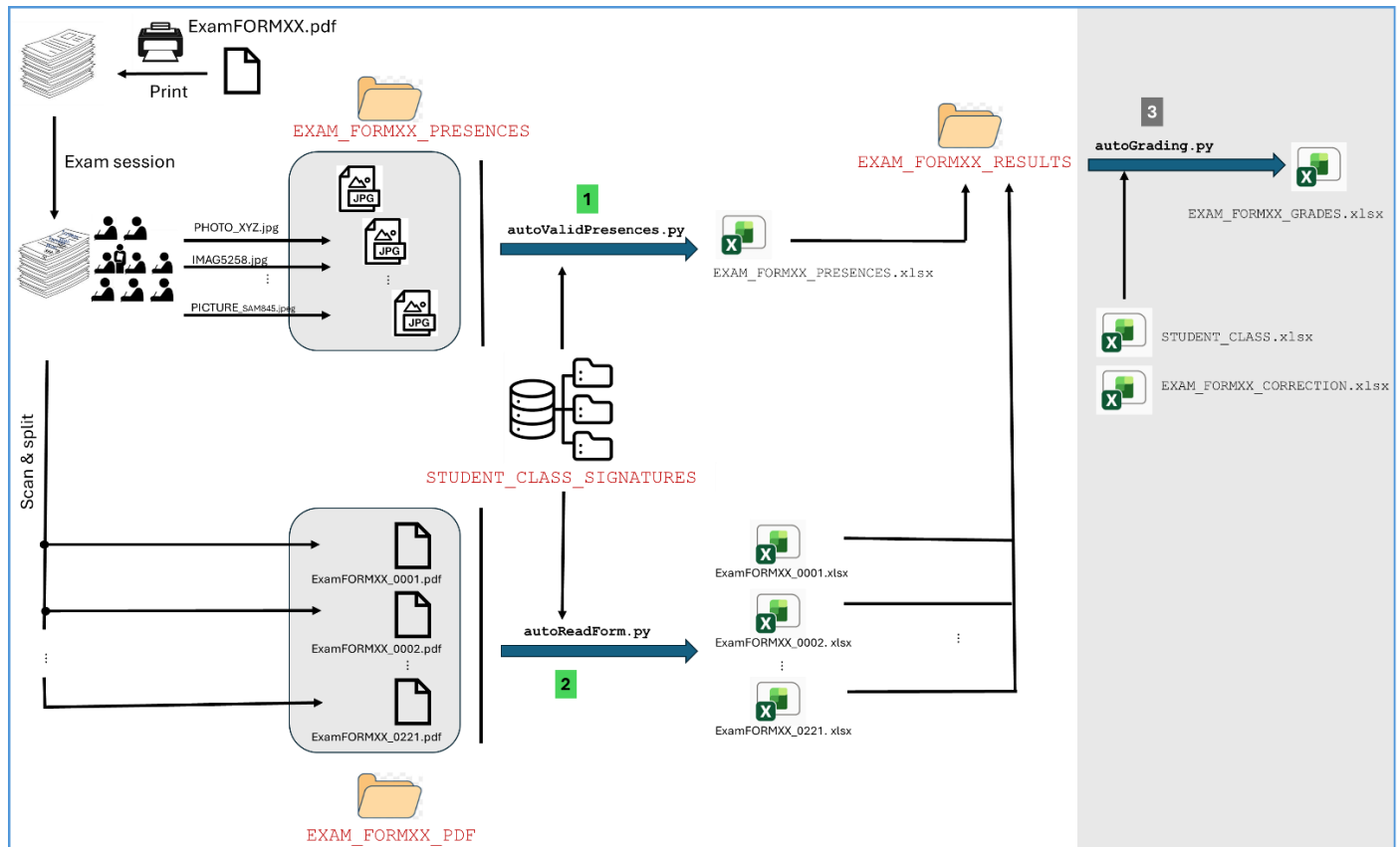


Figure 1. Automatic analysis, general structure.

Programs 1 and 2 are to be completed in this project; Program 3 is not part of the project

3.3 PROGRAM 1 : Attendance validation

Main function:

```
autoValidPresences(EXAM_FORMXX_PRESENCES, STUDENT_CLASS_SIGNATURES, EXAM_FORMXX_RESULTS)
```

It generates the file **EXAM_FORMXX_PRESENCES.xlsx**. This file contains three columns named respectively **imageName**, **studentID_grid**, and **studentID_signature**.

- Column A: the name of the processed image, including its file extension
- Column B: the student ID extracted from the graphical grid
- Column C: the student ID extracted from the signature sub-image

To do this, the function **autoValidPresences** creates the file **EXAM_FORMXX_PRESENCES.xlsx** and calls the sub-function.

```
autoValidID(filename.jpg, STUDENT_CLASS_SIGNATURES, EXAM_FORMXX_PRESENCES.xlsx, EXAM_FORMXX_RESULTS)
```

to process every image **filename.jpg** of the folder **EXAM_FORMXX_PRESENCES** and fill in the file **EXAM_FORMXX_PRESENCES.xlsx** for this particular image.

The sub-function **autoValidID**

- Extract the StudentID from the grid → **StudentID_grid**

- Extract the signature sub-image
- Compare the signature with the signature database (`STUDENT_CLASS_SIGNATURES`) and infer the corresponding studentID→ StudentID_bitmap
- Append a new row to the file `EXAM_FORMXX_PRESENCE.xlsx` filling the three columns accordingly.

Thus, the file `EXAM_FORMXX_PRESENCES.xlsx` provides an overall attendance status for the exam. If the student is correctly identified, columns B and C contain the same StudentID. If the signature is not recognized, column C is left empty. In case of identity fraud, columns B and C contain different StudentIDs. The professor must perform a manual verification whenever columns B and C differ, or one of the two columns is empty.

file will provide a general attendance status for the exam. If a student is correctly identified, columns 1 and 4 will contain the same StudentID; if the signature is not recognized, column 4 will be empty; if identity theft is suspected, columns 1 and 4 will be different, and the teacher will need to perform a verification.

`EXAM_FORMXX_PRESENCES.xlsx` will be saved in the folder `EXAM_FORMXX_RESULTS`.

3.4 PROGRAM 2: automatic form reading

Main function:

```
autoReadForm (EXAM_FORMXX_PDF, STUDENT_CLASS_SIGNATURES, EXAM_FORMXX_RESULTS)
```

This function lists all the PDF forms in the `EXAM_FORMXX_PDF` directory. It calls the `autoReadFormID` function on each PDF file to generate the corresponding Excel file based on the automatic reading performed.

```
autoReadFormID (EXAM_FORMXX_abcd.pdf, STUDENT_CLASS_SIGNATURES, EXAM_FORMXX_RESULTS)
```

The Excel file generated by `autoReadFormID` has the same name as the input PDF, for example: `EXAM_FORMXX_0001.xlsx`, for the file `EXAM_FORMXX_0001.pdf`. The generated Excel files are all saved in the `EXAM_FORMXX_RESULTS` folder.

Sub-function `autoReadFormID`:

- Open the file `EXAM_FORMXX_abcd.pdf` and convert each page into an image to be processed.
- Interpret page 1 to fill in tab `PAGE-01` of the Excel file `EXAM_FORMXX_abcd.xlsx`.
- Interpret the exam pages (p5→end) to fill in the `EXAM` tab of `EXAM_FORMXX_abcd.xlsx`.
- Verify that all the cryptograms at the bottom of the page are identical and fill in the corresponding cell in sheet `PAGE-01`.

Additional information:

- Signature validation is here done using the PDF form.
- The STUDENT ID and group number are read from the grid, not from handwritten information.
- The cryptogram is the small graphic at the bottom of the page, for example.



It must be the same on all pages.

- Reading all the general exam information provided on page 1 allows for authentication of both the student and the form.

The figure below shows the structure of the generated Excel files.

	A	B		A	B	C	D	E	F	G	H	I	J	K	L
1	Module	IS.2703		QUESTION	CHOIX A	CHOIX B	CHOIX C	CHOIX D	CHOIX E	CHOIX F	CHOIX G	CHOIX H	MANTISSE	EXPOSANT	UNITE
2	Professor	GASTON		1		1									
3	Date	04/04/2026		2			1								
4	Code	S2-02-G2		3									3,250000	-1	
5	Notes de cours	0		4	1										
6	Notes manuscrites	1		5									0,870000	0	
7	Ordinateur portable	0		6									0,650000	0	
8	Calculatrice	1		7		1									
9	Feuilles brouillon	2		8	1										
10	Note maximale	20		9				1							
11	Note pour valider	10		10									188,750000	0	
12				11	1			1		1					
13	Prénom	LINA		12											
14	Nom	MOREL													
15	Validation signature	1													
16	Group	G02B													
17	STUDENT ID	48271													
18	Validation cryptogramme	1													

PAGE-01

EXAM

Figure 2. Structure of the generated .xlsx file from automatic processing. The columns depend on the exam form being processed

3.5 PROGRAM 3 : Fusion (not required)

The files generated by programs 1 and 2, `EXAM_FORMXX_PRESENCES.xlsx` and `EXAM_FORMXX_abcd.xlsx`, saved in the `EXAM_FORMXX_RESULTS` directory, will be combined with the exam correction Excel file (created by the teacher) to generate a single Excel file of results. For each student, the following information will be available:

- Confirmation of attendance
- Authentication of his signature
- Authentication of the form
- The grade derived from the answers

DeepForm plans to implement this feature in phase 2 of the project, once programs 1 and 2 have been validated. Therefore, Isep students will not need to complete this step.

3.6 MAIN program

The main program defines:

- Exam name : `ExamFORMXX`
- Signature folder : `STUDENT_CLASS_SIGNATURES`

It automatically deduces:

- PDF folder: `ExamFORMXX_PDF`
- Image folder (1st pages): `EXAM_FORMXX_PRESENCES`
- Results folder: `ExamFORMXX_RESULTS`

It creates the directory `ExamFORMXX_RESULTS`.

It runs `autoValidPresences` and `autoReadForm`.

4 Instructions

4.1 Programming

This project is an application of computer vision in which you will need to recognize graphical elements, printed text, and handwritten text. Your programs must comply with the following guidelines:

- **Graphical elements** must be processed using low-level functions, i.e., methods covered in class such as filtering, Hough transform, mathematical morphology, image rotation, etc. The use of higher-level functions is prohibited, for example automatic rectangle detection, checkbox detection, etc. It is up to you to design and implement your own methods.
- **Handwritten or printed text** may be processed using higher-level functions. For example, you may reuse neural networks for handwritten digit recognition and fine-tune them if necessary. However, you must

choose appropriate tools that are properly adapted to the problem, be able to justify your choices and parameters, and explain the underlying methods and architectures.

4.2 Methods

4.2.1 Databases

This project is a typical computer vision application, involving preprocessing, segmentation, and recognition stages. The segmentation and recognition steps will require defining a number of parameters, such as object dimensions, similarity thresholds, etc., or training a neural network. It is essential to develop the system using a training dataset and to optimize the hyperparameters using a validation dataset. The generalization ability of the algorithms on unseen data will be evaluated using a test dataset. These three datasets must be strictly separate.

This is a general methodology that must be followed in any supervised recognition problem (i.e., designed using annotated data).

4.2.2 Quantitative evaluation

The provided dataset contains ground truth data. Therefore, you will be able to perform quantitative evaluations to objectively assess your programs at multiple levels. It is up to you to choose the evaluation criteria, but it is recommended to take inspiration from the information given in the challenge (Section 6).

Recommendation: use cross-validation.

5 Design

You will design your project in the framework Vocareum. You must comply with the specifications provided for the functions in Section 3. Your main program must run on the data provided for the challenge without crashing. The only modifications allowed for the challenge will be the definitions related to the input data in the main program.

6 Challenge

The teachers will provide a new dataset (in accordance with what is described in Section 3.1). This data will be entirely new and may include identity fraud, page swaps between students, forged signatures, etc.

Only programs that comply with the instructions in Section 4.1 and that run fully automatically without any manual correction will be eligible for the challenge. The results produced by your programs will be evaluated according to four criteria.

6.1 Signature authentication performance

- Authentication of the signature extracted from photos (Program 1)
- Reading the StudentID from photos (Program 1)
- Authentication of the signature extracted from the first page of the PDF (Program 2): line 15 of the `PAGE-01` sheet.

6.2 Printed information reading performance

- From page 1 of the PDF: lines 1–4 and 9–11 of the `PAGE-01` sheet

6.3 Handwritten information reading performance

- Lignes 13 et 14 of tab `PAGE-01`
- Column MANTISSA EXPONENT of tab `EXAM`

6.4 Graphical information reading performance

- Lines 5 -9 and 16-18 of tab PAGE-01
- Column CHOIX of tab EXAM.

Execution time will also be taken into account.

For each category, bonus points will be awarded to the gold, silver, and bronze medalists. Additional rewards may be granted to gold medalists...

7 Deliverables

- Codes
- Report
- Defense

7.1 Report

This report must be a well-written and rigorous document, prepared in the style of a scientific paper. It may be written in English or French. It must include the following elements:

- Problem statement
- State of the art (brief). References should be cited using numbers [1], [2], ...
- Description of the methods: note that this does not involve copying code, but rather formalizing the applied processing using equations. No code or pseudocode should appear in the report. You should begin by presenting the main stages of the program (preprocessing, segmentation, etc.) using a functional diagram. For each stage, introduce the methods and variables in text and formalize them mathematically with equations. Clearly highlight the program parameters. Illustrate intermediate results with images. Neural network architectures must be described using a diagram, along with all key training elements (datasets, loss functions, data augmentation, etc.).
- Experimental results and discussion
- Conclusion and future work

Note: It is allowed to reuse existing code (provided the instructions in Section 4.1 are followed), but sources must be cited accurately, even if only part of the final program is concerned. Otherwise, **the work will be considered plagiarism and will be graded 0**. Reused code must be fully understood, explained, discussed, and possibly improved. The final grade will take into account your personal contribution.

7.2 Defense

The defense will last 40 minutes per team, with 25 minutes for the presentation and 15 minutes for questions. Please strictly adhere to this timing; it will not be possible to exceed the allotted presentation time. The defense may be conducted in either English or French.

Prepare a PowerPoint presentation. The PowerPoint presentation should be created in the same spirit as the report.

7.3 Codes

The framework is Vocareum.

8 Evaluation

Skills assessed

- Ability to design a computer vision system addressing a given problem. Relevance of the chosen methods. Personal contribution.
- System-level understanding (functional diagram).
- Ability to explain and formalize the proposed methods (report, presentation).
- Quality of writing and oral presentation: structure, rigor, spelling, clarity, etc.
- Experimental study: quantitative evaluation, critical analysis.
- Compliance with the methodology and specifications: training, testing, data format and program specifications.
- Quality of programming: modular, well-structured code, clearly defined parameters (not hard-coded), well-commented code, etc.
- Quality of results and critical thinking.

The final grade will be based on this skill assessment and will take into account the results of the challenge. A bonus will be awarded to teams that reach the podium in at least one of the four challenge tracks!